Chapter 1: Introduction to Python

Python Programming Study Guide Series

OVERVIEW & CORE CONCEPTS

Python is an open-source, high-level, interpreted programming language created by **Guido van Rossum** and released in 1991. Python emphasizes code readability and programmer productivity, allowing developers to express concepts in fewer lines of code than C++ or Java.

Key Characteristics

- **Interpreted:** Code is executed line-by-line without pre-compilation.
- **Dynamically Typed:** Variables do not require explicit type declaration.
- **Cross-Platform:** Runs seamlessly across Windows, macOS, and Linux.

Major Domains

- **Web Development:** Frameworks like Django and Flask.
- **Data Science & AI:** NumPy, Pandas, PyTorch, and TensorFlow.
- **Automation:** Scripting, system administration, and scraping.

FIRST PROGRAM: HELLO WORLD

In Python, the `print()` function outputs text directly to the console. Unlike many traditional languages, Python does not require boilerplate classes or semicolon line terminators.

```
# Chapter 1: First Python Program
print("Hello, World!")
```

Key Takeaway

Python code structure relies heavily on indentation (whitespace) to group statements. Standard Python code uses 4 spaces per indentation level instead of curly braces `{}`.